

Natural Products in Drug Design

Aim: The aim of this practical work was to perform computational structural similarity analysis between approved drug molecules and natural product compounds using cheminformatics techniques to investigate structural relationships relevant to drug discovery and repurposing. The main objective involves the identification of common molecular scaffolds shared between compounds through Maximum Common Substructure (MCS) analysis that leads to assessing structural similarity within and across datasets. This practical also aimed to demonstrate the use of computational structural bioinformatics tools for large-scale molecular comparison, data visualization, and similarity-based compound prioritization.

Task Description: The practical involved implementing a molecular similarity workflow in Python using the RDKit cheminformatics library. Two molecular datasets were provided in CSV format that comprises of approved drug compounds and natural product molecules represented using SMILES notation. The workflow required initial preprocessing of the datasets such as standardizing molecular structure representations, and calculating pairwise structural similarity using Maximum Common Substructure analysis. Similarity criteria involves the quantification by comparing the number of shared atoms in the common substructure relative to the total atoms present in the compared molecules.

Three major comparisons were performed as part of the task: structural comparison among approved drug molecules, structural comparison among natural product compounds, and crosscomparison between approved drugs and natural products. Overall, there were large-scale computational analysis involving 46,225 drug-versus-drug comparisons, 996,004 natural product-versus-natural product comparisons, and 214,570 drug-versus-natural product comparisons. Due to computational complexity of exhaustive MCS calculations, parallel computing technique using Python multiprocessing to accelerate execution was employed. The workflow also included exporting similarity data into structured CSV tables and generating heatmap visualizations to illustrate clustering and structural relationships across datasets.

Results and Discussion

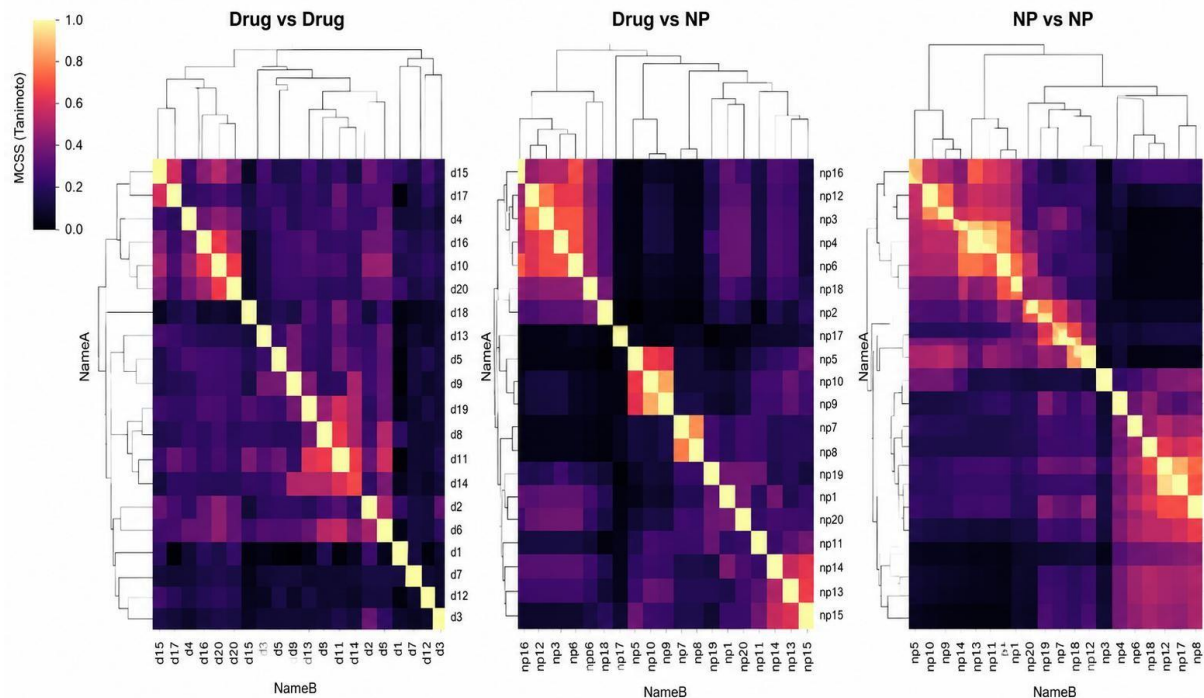


Figure 6. Comparative clustered heatmap visualization of structural similarity analysis between approved drugs (Drug vs Drug), approved drugs and natural products (Drug vs NP), and natural products (NP vs NP) based on Maximum Common Substructure (MCSS) similarity scores

Table 4: cross-comparison between approved drugs and natural products

Na me A	Na me B	CmpdA	CmpdB	A #	B #	MCS	MCSAtto-mCount	MC SS
d1	np1	CC(=O)NCCC S(O)(=O)=O	CN1C=NC(C[C@H](N)C(O)=O)=C1	1	1	[#6&!R]- &!@[#6&!R]= &!@[#8&!R]	3	0.15
d1	np2	CC(=O)NCCC S(O)(=O)=O	NCCCN	1	5	[#7&!R]- &!@[#6&!R]- &!@[#6&!R]- &!@[#6&!R]	4	0.33 3333
d1	np3	CC(=O)NCCC S(O)(=O)=O	CCC(=O)C(O)=O	1	7	[#6&!R]- &!@[#6&!R]- &!@[#6&!R]	3	0.2
d1	np4	CC(=O)NCCC S(O)(=O)=O	CCC(O)C(O)=O	1	7	[#6&!R]- &!@[#6&!R]- &!@[#6&!R]	3	0.2
d1	np5	CC(=O)NCCC S(O)(=O)=O	[H][C@@]12CCC(=O)[C@@]1(C)CC[C@]1([H])C3 =C(CC[C@@]21[H])C=C(O)C(OC)=C3	1	2	[#6&!R]	1	0.03 125

Table 5: Structural comparison among approved drug molecules (Drug vs drug)

Na	CmpdA	CmpdB	A	MCS	MCSA	tomCo	MC
Na		B					SS
me			#		unt		
me A			#				
B							
d1	d1	CC(=O)NCCCS(O)(=O)=O	CC(=O)NCCCS(O)(=O)=O		[#6&!R]-	11	1
			1		1 &!@[#6&!R](=&!@[#8		
					1 &!R])-&!@[#7&!R]&!@[#6&!R]-		
					1 &!@[#6&!R]-		
					&!@[#6&!R]&!@[#16&!R](&!@[#8&!R])(=&!@[#8		
					&!R])=&!@[#8&!R]		
d1	d2	CC(=O)NCCCS(O)(=O)=O	CC(=O)NC1=CC=C(O)C=C1	1 1	[#6&!R]-	4	0.2
				1 1	&!@[#6&!R](=&!@[#8		222
					&!R])-&!@[#7&!R]		&!R])-
d2	d1	CC(=O)NC1=CC=C(O)C=C1	CC(=O)NCCCS(O)(=O)=O	1 1	[#6&!R]-	4	
				1 1	&!@[#6&!R](=&!@[#8		
					&!R])-&!@[#7&!R]		
d1	d3	CC(=O)NCCCS(O)(=O)=O	CCN1N=NN(CCN2CCC(COC)(	1 3	[#6&!R]-	4	
			CC2)N(C(=O)CC)C2=CC=CC=	1 0	&!@[#6&!R](=&!@[#8		
			C2)C1=O		&!R])-&!@[#7&!R]		
					22		
					0.2		
						081	222
d3	d1	CCN1N=NN(CCN2CCC(COC)(	CC(=O)NCCCS(O)(=O)=O	3 1	[#6&!R]-	4	0.1
	08						
	0.1						
		CC2)N(C(=O)CC)C2=CC=CC=		0 1	&!@[#6&!R](=&!@[#8		081
		C2)C1=O			&!R])-&!@[#7&!R]		08

Table 6: Structural comparison among Natural products (Np vs Np)

NA	NA	CMPDA	CMPDB	A	B	MCS	MCSATO	MC
ME	ME			#	#		MCOUNT	SS
A	B							
NP1	np1	CN1C=NC(C[C@H]	CN1C=NC(C[C@H]	1 1	[#6&!R]-		12	1
		(N)C(O)=O)=C1	(N)C(O)=O)=C1	2 2	&!@[#7]1:&@[#6]:&@[#7]:&@[#6](:			
					&@[#6]:&@[#6]1)-&!@[#6&!R]-			
					&!@[#6&!R](-&!@[#7&!R])-			
					&!@[#6&!R](-			
					&!@[#8&!R])=&!@[#8&!R]			
NP1	np2	CN1C=NC(C[C@H]	NCCCN	1 5	[#7&!R]-&!@[#6&!R]-&!@[#6&!R]		3	0.21
		(N)C(O)=O)=C1		2				4286
NP2	np1	NCCCN	CN1C=NC(C[C@H]	5 1	[#7&!R]-&!@[#6&!R]-&!@[#6&!R]		3	0.21
			(N)C(O)=O)=C1	2				4286
NP1	np3	CN1C=NC(C[C@H]	CCC(=O)C(O)=O	1 7	[#6&!R]-&!@[#6&!R]-&!@[#6&!R](-		5	0.35
		(N)C(O)=O)=C1		2	&!@[#8&!R])=&!@[#8&!R]			7143

NP3	np1	CCC(=O)C(O)=O	CN1C=NC(C[C@H]	7	1	[#6&!R]-
		&!@[#6&!R]-&!@[#6&!R](-	5	0.35 (N)C(O)=O)=C1		2
		&!@[#8&!R])=&!@[#8&!R]	7143			

The analysis successfully identified structural similarity patterns across all molecular comparisons. Drug-versus-drug analysis revealed several clusters of compounds with high structural similarity demonstrates shared chemical scaffolds and related therapeutic classes. Self-comparisons produced similarity scores of 1.0 indicating correct implementation of the similarity metric. The results indicates that natural product comparisons leads to a wider degree of structural diversity, although multiple highly similar clusters were observed that indicates conserved scaffold relationships among natural compounds.

Furthermore, cross-comparison between approved drugs and natural products identified several strong structural matches that indicates higher similarity scores exceeding 0.90 due to substantial overlap between known drug scaffolds and naturally occurring compounds. The results of molecular similarity analysis indicates that this is an effective computational approach for identifying structurally related compounds in drug discovery. The clustering observed in the drug dataset indicates shared chemical scaffolds among approved therapeutics and the natural product dataset showed both structural diversity and conserved molecular frameworks that is comparable with previous findings on natural product chemical space (Newman and Cragg, 2020). The high similarity scores observed between certain approved drugs and natural products suggest potential opportunities for drug repurposing as similarity among compounds exhibit related biological activities (Cereto-Massagué et al., 2015). However, structural similarity alone is insufficient to confirm therapeutic potential due to various other factors insolvent and biological activity also depends on binding affinity, pharmacokinetics, and toxicity (Ekins et al., 2019). Therefore, additional analyses such as molecular docking and ADMET prediction would be required to validate these findings. Overall, the practical successfully demonstrated the application of computational structural bioinformatics methods for large-scale molecular similarity analysis in a drug discovery context.